13 The diameter of the receiving dish of a telescope is 5.1 m. It receives light of wavelength 0.40 μm from two point sources at a distance of 2.8×10^{25} m away.

What is the minimum separation of the two points sources so that the light from them can be resolved?

A 1.1×10^{18} m

B 2.2×10^{18} m

C 1.8×10^{32} m

D 3.6×10^{32} m

Ans: B

The minimum angular separation for two point sources to be resolved is: $\theta_{min} \approx \frac{\lambda}{b}$ where *b* is the diameter of the receiving dish.

minimum angle:

$$\theta_{min} \approx \frac{0.40 \times 10^{-6}}{5.1} = 7.843 \times 10^{-8} \text{ rad}$$

minimum separation:

For small angles, the separation *x* at distance *D* is: $x = \theta_{min} \times D$

$$x = 7.843 \times 10^{-8} \times 2.8 \times 10^{25} \approx 2.2 \times 10^{18} \text{ m}$$

14 An ideal gas has thermodynamic temperature *T* and its molecules has root-mean-square